A Compact, Portable Spectrometer for NMR Measurements in Undergraduate Education Labs

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***Abstract*—Compact and affordable quantum computing hardware is needed to provide secondary-school and undergraduate students with hands-on experience in physics and engineering. We present Quorum-1, a low-cost, open-source, single-qubit Earth-field nuclear magnetic resonance platform designed for hands-on quantum education and adaptable to both spectroscopy and gradient-coil magnetic resonance imaging demonstrations. Operating in Earth's magnetic field, the platform supports instructional NMR measurements and proof-of-concept experiments relevant to quantum control and magnetic resonance imaging. We also evaluate practical implementation considerations, particularly cost, and find that the device can be assembled using materials totaling approximately $500. These results position Quorum-1 as an accessible hardware platform for quantum education and related magnetic resonance demonstrations.**

I. Introduction

Modern scientific education emphasizes the importance of exposing students to current technologies. And to meet the demand, there is a growing need for quantum technology curricula for the emerging quantum market [1]. While cloud-based platforms for quantum computing resources like IBM Quantum [2] and Amazon Braket [3] make quantum information science (QIS) instruction and research readily available, these online resources do not address the need for hands-on quantum hardware instructional materials. Even the few commercially available desktop quantum computing devices [4], [5] do not give access to the underlying electronics. Their price tag also limits which institutions can experiment with them.

There is a market opening for quantum computing devices which can provide hands-on training in the electronics and pulse engineering skills that employers are seeking in the quantum workforce [6], [7]. Such a device would be useful to the needs of undergraduate quantum programs at the bachelor and associate degree level [8] as well as a supplement to introductory quantum material in high school curricula [9] and for motivated independent learners. With these challenges in mind, we present an open-source hardware package that offers a tangible and tactile experience with quantum devices. Full circuit schematics, parts list, and control software are available online [10] along with an accompanying series of video tutorials [11].

The inspiration for our setup was taken from previous low-cost NMR devices available on the market such as the Terranova NMR spectrometer [12] and explicitly builds on the inexpensive portable spectrometer developed by Michal [13]. Although our primary goal was to improve and adapt Michal's system for use in liquid-phase NMR quantum computing , it is still possible to make use of our system for other NMR spectroscopy use cases such as the traditional Earth's magnetic field physics experiment [14]. Additionally, our design can be adapted to perform nuclear magnetic resonance imaging (MRI) with the addition of a flexible Golay coil assembly [15].

II. Background

A. Liquid-Phase Nuclear Magnetic Resonance

Liquid-phase NMR was one of the earliest platforms used to demonstrate quantum information processing [16]. Unlike many quantum computing platforms that rely on cryogenic operation to prepare nearly pure single-system states, liquid-state NMR operates above the solvent's freezing point on large molecular ensembles. As a result, computation uses weakly polarized, highly mixed spin states, and measurements reflect ensemble-averaged magnetization rather than projective readout of individual qubits. In an external magnetic field, spin-1/2 nuclei form two-level systems through Zeeman splitting, providing a natural qubit basis. Differences in resonance frequency permit selective control of individual spins, while spin-spin couplings enable multi-qubit operations. The ensemble state is detected inductively through the signal generated by its collective magnetization in a pickup coil.

B. Nuclear Spin Precession as a Single-Qubit Model

A proton in a magnetic field has two spin basis states, commonly labeled $|0\rangle$ and $|1\rangle$, corresponding to spin aligned or anti-aligned with the static magnetic field. A general single-spin state can be written as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \tag{1}$$

where α and β are complex amplitudes. The Bloch sphere representation of this state is directly connected to the classical NMR picture of magnetization precessing about the magnetic field.

At room temperature and Earth-field strength, the water sample is not initialized as a pure $|0\rangle$ state. Instead, it is a large thermal ensemble with a very small population imbalance between the two spin states. In the high-temperature approximation, the density matrix can be written approximately as

$$\rho \approx \frac{I}{2} + \epsilon \frac{\sigma_z}{2} \tag{2}$$

where $I/2$ is an unobservable maximally mixed background and ϵ is the small thermal polarization. The measured NMR signal comes from the deviation magnetization, which behaves under RF pulses like a controllable Bloch vector [17], [18].

C. Challenges with LPNMR Quantum Computation

While prevalent in the early days of quantum computing research, LPNMR was found to be unsuitable to the requirements of industrial scale computations and largely abandoned in favor of more promising architectures. Scaling LPNMRQC is difficult because are the computational space is bound to a single molecule in a bulk isotropic liquid, requiring larger molecules with individually addressable atomic nuclei and limiting qudit connectivity [19].

Although liquid-state NMR quantum computing was the subject of considerable debate regarding the role of entanglement in highly mixed pseudopure states, the current understanding is well established. The nuclear spin dynamics of molecules in a room-temperature liquid are fully governed by quantum mechanics, and coherent radio-frequency pulses implement unitary operations on the spin Hilbert space. However, because the thermal polarization is extremely small, the complete density operator of the ensemble remains highly mixed and is generally separable under conventional high-temperature operating conditions. Consequently, while liquid-state NMR provides genuine quantum coherent control and accurately realizes quantum dynamics, conventional bulk-ensemble NMR does not provide a scalable architecture for universal quantum computation due to the exponentially decreasing signal associated with pseudopure-state preparation. Even though liquid-state NMR is no longer regarded as a practical architecture for scalable quantum computing, its mature instrumentation, long coherence times, and exceptional degree of coherent control continue to make it an attractive platform for quantum information experiments and education [19]–[21].

III. Design

The present system (Figure 1) builds on the design reported by Michal [13] while incorporating improvements enabled by advances in commercially available electronics. The control platform was migrated from a 5 V Arduino-based system to a 3.3 V Raspberry Pi Pico RP2040 [22], hereafter referred to as the Pico. Relative to the Arduino platform, the Pico offers greater processing capability, increased memory capacity, and lower retail cost. These improvements enable finer frequency resolution and greater control accuracy. Migration to the 3.3 V platform also required several hardware modifications to ensure compatibility with the lower logic-level voltage. In particular, a voltage divider was added at the output of the new relays to keep the analog-to-digital converter (ADC) input within its allowable range. The control software was rewritten from the Arduino development environment to native C using the Raspberry Pi Pico SDK. C was selected over MicroPython to maximize execution speed.

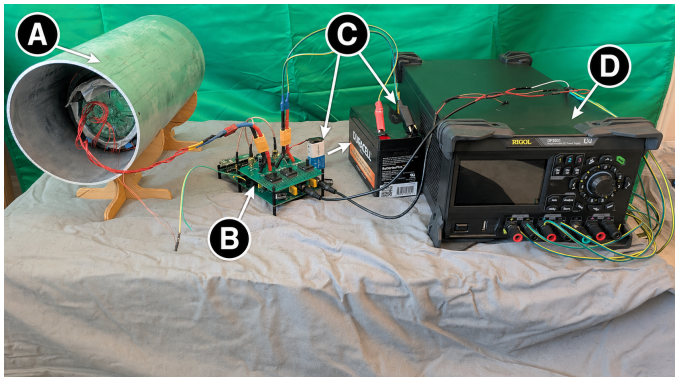


Fig. 1: Photograph of the complete Quorum-1 NMR experimental setup. The apparatus is comprised of (A) the shielded sample and coil assembly, (B) the RF and control electronics board, (C) the auxiliary battery supplies, including a 9 V battery and a 12 V sealed lead-acid battery, and (D) the programmable DC power supply.

The Quorum-1 PCB was designed primarily with through-hole components to simplify assembly and improve accessibility for educational use. To reduce wear on sensitive components and facilitate maintenance, major subsystems are connected via removable connectors. Several features were incorporated to reduce power-supply line noise and support systematic troubleshooting, including a continuous ground plane, a dedicated split layer for the +15 V and -15 V rails, and additional through-hole positions at the power input for optional low-pass capacitors.

We have adopted many of the suggestions put forth by van der Veeken [23] in his comprehensive study of Michal's original design specifically in the design of the transmit/receive and polarization coils. For further optimization, jumpers were placed between operational-amplifier stages, and probe points were added after each primary component and at relay interfaces to aid fault isolation. Signal transmission between boards is implemented using coaxial SMA cables to reduce susceptibility to external interference. Additional choices, including polarity-agnostic audio jack inputs for the ±15 V supplies, female headers for easier component replacement, and protected power-input connections that help ensure correct voltage polarity, further improve system robustness and maintainability. A full photo of the control board setup can be seen in Figure 2.

A. PCB Operation

The signal-generation pipeline consists of six operational-amplifier stages. In the first stage, two interleaved signals

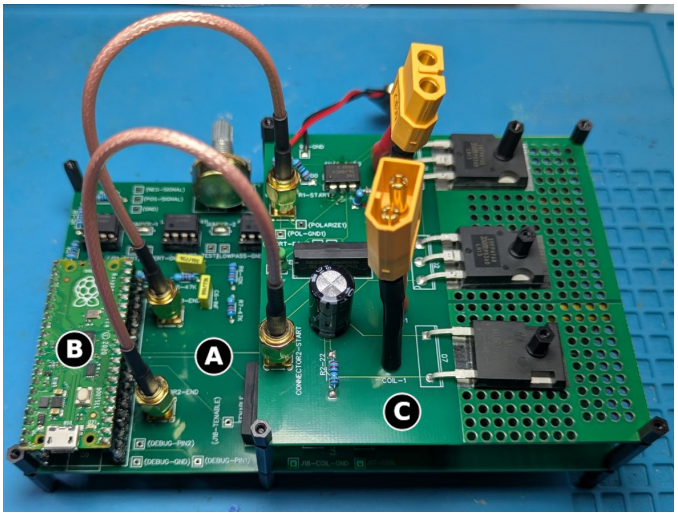


Fig. 2: Close-up photograph of the control electronics assembly. The assembly consists of (A) the main control board, (B) the Raspberry Pi Pico microcontroller, and (C) the polarization board.

from two general-purpose input/output (GPIO) pins on the Raspberry Pi Pico are used to enable arbitrary pulse generation. These signals are combined in a differential amplifier configuration to produce a bidirectional square wave. In the second stage, the waveform passes through a low-pass filter to smooth the square wave into an approximate sinusoid. The signal is then sent to an amplification stage, with gain set by a potentiometer mechanically controlled by the Pico via a servo motor. This configuration enables fine amplitude control for power-Rabi experiments. The amplified signal is then applied to the antenna for transmission into the sample. Three Pico-controlled relays are located at the antenna interface. These relays allow the coil to be isolated from either half of the device and grounded prior to operation, thereby preventing residual resonance buildup.

After excitation, the free induction decay (FID) is received by the same transceiver antenna and routed to the fourth operational-amplifier stage. While band-pass filtering would ideally be performed immediately after reception, the signal amplitude at this point is too low for effective filtering, necessitating initial amplification. The amplification stage is configured for a gain of approximately 1000×. Once amplified, the signal is band-pass filtered around 2100 Hz. The filter design is relatively permissive and has been found to operate effectively with a bandwidth of approximately 100 Hz. In the final stage, the signal is shifted upward in voltage to accommodate the Raspberry Pi Pico analog-to-digital converter (ADC), which cannot directly sample bipolar signals. The Pico ADC operates at 500,000 samples/s, although memory limitations restrict each acquisition to 50,000 samples. The acquired data are then transmitted to a host PC via USB for fast Fourier transform based processing and further analysis.

The second primary board controls the polarization hardware. Similar to the transceiver antenna, the polarization coil is connected to a relay that grounds the coil when inactive. Because the relay is arranged in parallel with the coil, improper timing can expose it to damaging current levels. To mitigate the risk of costly relay failure, a 500 mA fuse was placed in series with the relay. The Raspberry Pi Pico drives an optocoupler that allows a 9 V battery signal to switch two parallel MOSFETs. These MOSFETs control the delivery of more than 10 A to the main polarization coil.

A key improvement over Michal's breadboard-based implementation is the use of a printed circuit board (PCB). The inclusion of plated through-holes and a copper plane improves heat dissipation from the MOSFETs, removing the need for supplementary heat sinks. A MUR3060 diode is also included for flyback-current suppression. Finally, the 900 µH polarization coil, 1 mF capacitor, and 22 Ω resistor form an overdamped network.

IV. Results and Discussion

We measured an ambient magnetic field of approximately 52 µT using a Google Pixel 8a smartphone magnetometer, consistent with the NOAA/WMMHR-2025 predicted total field at the measurement location (52.83 µT for 2026) [24].

Upon completing the revised design, we evaluated both functional performance and practical feasibility. Assembly trials showed that a single operator could complete the build in under four hours [11]. Assembly time will vary with the builder's experience in electronics and troubleshooting; however, the simplified architecture reduces setup and integration efforts.

A. Power Rabi

Power-Rabi measurements provide a convenient way to calibrate coherent spin rotations by varying the amplitude of the applied RF field while keeping the pulse duration fixed. In the rotating frame, an on-resonance RF magnetic field $(B_{1,\mathrm{eff}})$ drives nutation of the nuclear magnetization through a flip angle

$$\theta = \gamma B_{1,\mathrm{eff}} \tau, \tag{3}$$

where (γ) is the proton gyromagnetic ratio and (τ) is the effective pulse duration [18], [25]. As the RF amplitude is increased, the spin ensemble is rotated through successive $(\pi/2)$, (π), $(3\pi/2)$, and (2π) rotations, producing an oscillatory measured response. For a positive-valued signal metric derived from the detected spectral magnitude, the corresponding idealized response may be written phenomenologically as

$$S!\left(B_{1,\mathrm{eff}}\right) = S_0 + A\sin^2!\left(\gamma B_{1,\mathrm{eff}}\tau + \phi\right), \tag{4}$$

where (S_0), (A), and (ϕ) account for the signal offset, scale, and phase of the measured response. In the language of quantum control, this calibration identifies the RF amplitudes needed to implement single-qubit rotations such as $(R_x(\theta))$ or $(R_y(\theta))$, making the power-Rabi experiment a natural bridge

between low-field NMR measurements and gate-based spin control [17], [20].

Traditional pulse length Rabi testing is challenging on this device because a typical excitation pulse contains only about four oscillation cycles, which severely limits the achievable frequency resolution. By contrast, digitizing the gain-control potentiometer allows fine adjustment of the transmitted signal amplitude, making power-Rabi measurements more suitable for this platform. In the present analysis (Figure 4), the measured response is quantified by the magnitude of the dominant non-DC FFT peak and plotted as a function of servo power setting. The resulting dependence is approximately oscillatory, so a weighted phenomenological fit of the form

$$y(P) = y_0 + A\sin^2(kP + \phi) \tag{5}$$

is overlaid to summarize the trend. Because the mapping between servo power setting and effective RF magnetic field has not yet been independently calibrated, this dashed curve should be interpreted as an empirical fit in control-setting space rather than as a direct first-principles prediction.

For reference, the idealized response of the system to the effective rotating-frame RF magnetic field may be written as

$$S(B_{1,\mathrm{eff}}) = A\,\sin^2(\gamma B_{1,\mathrm{eff}}\tau) \tag{6}$$

where $S(B_{1,\mathrm{eff}})$ is the predicted measured signal metric, A is an overall scale factor fit to the measured amplitude, γ is the proton gyromagnetic ratio $\left(\gamma = 2\pi \cdot 42.577 \times 10^6\ \mathrm{rad\,s^{-1}\,T^{-1}}\right)$, $B_{1,\mathrm{eff}}$ is the effective rotating-frame RF magnetic field amplitude at the sample, and τ is the total effective drive duration. In the absence of an independent calibration relating servo power setting to $B_{1,\mathrm{eff}}$, Eq. (6) is presented as a qualitative physical reference rather than the exact function used for the dashed overlay in Fig. 4.

B. Magnetic Resonance Imaging

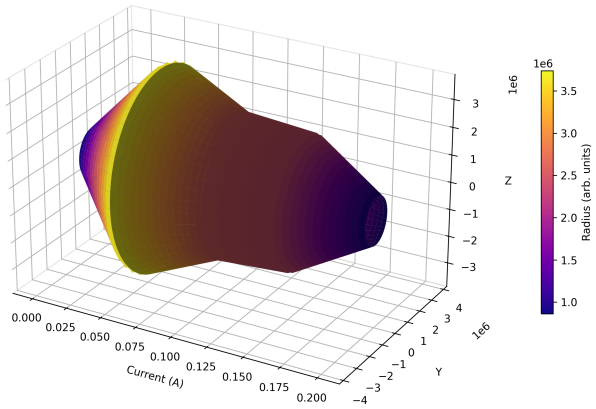
Through the addition of a flexible Golay coil and automated frequency sweeping, Quorum-1 is capable of producing proof-of-concept one-dimensional MRI reconstruction.

The gradient coil was designed using CoilGen [29]. Using a geometrical script, we flattened the trajectories and loaded them into both FreeCAD [30] and pyCAD [31]. After some cleanup we were left with a Golay coil which could be manufactured on a flexible PCB by the company PCBWay. This gradient coil was designed to fit between the transceiver and polarization coils. Through experimentation we found that 200 mA on the coils was enough to fully cancel out the Earth's field of about 50 µT. We were able to run 5 different values and found qualitatively similar shapes between the MRI image and the expected ground truth object (Figure 3). This way the flexible PCB can fit snugly between the polarization and the transceiver coils. We tested the field gradient using and an Bosch Sensortech BNO055 magnetometer [32] and a Meta Quest 2 system for three-dimensional visualization [33].

Frequency sweeping for spatial encoding was automated using a motor-driven potentiometer coupled to the RF tuning knob. A varistor may be used to the same effect for full solid state control.



(a) Photograph of the soy sauce bottle used as the test object. Note that the bottle was imaged in a horizontal orientation.



(b) Axisymmetric reconstruction obtained from the processed gradient-encoded measurements.

Fig. 3: Reference photograph (a) and corresponding preliminary axisymmetric reconstruction (b) for the soy-sauce-bottle demonstration. The reconstructed shape captures the object's gross bottle-like profile and is shown for qualitative comparison with the physical test object.

C. Value Engineering

Cost was treated as a primary design constraint throughout the project. Components and fabrication choices were selected to limit total expense while maintaining mechanical robustness and required functionality. The resulting device reflects a compromise between accessibility, cost, and capability.

An essential design constraint was to ensure the entire device could be constructed with easily obtainable components that could be purchased either at a hardware or electronics

TABLE I: Cost comparison of selected NMR-based educational and small-scale quantum computing platforms in the U.S. market.

Device	Platform	Qubits	Price (USD)
SpinQuorum Quorum 1	Earth-field NMR	1	$500
TeachSpin Quantum Control	NMR / quantum control	1	$5,750 [26]
SpinQ Gemini Mini	NMR quantum computer	2	$8,700 [27]
TeachSpin Earth's Field NMR with Gradient Field Coils	Earth-field NMR	1	$9,400 [26]
SpinQ Gemini Lab	NMR quantum computer	1–2	>$10,000 [28]
SpinQ Gemini	NMR quantum computer	2	$40,000 [27]
SpinQ Triangulum	NMR quantum computer	3	$58,000 [27]

Note: The SpinQuorum Quorum 1 pricing is the estimated cost of components and materials per device, not a commercial retail sale price. A detailed price breakdown can be found in appendix Table II. TeachSpin lists its posted prices as estimates because package options vary. The SpinQ Gemini Mini, Gemini, and Triangulum prices are approximate publicly reported 2022 prices from press coverage. SpinQ does not list prices for its products publicly with the exception of an approximate range given for SpinQ Gemini Lab on their website.

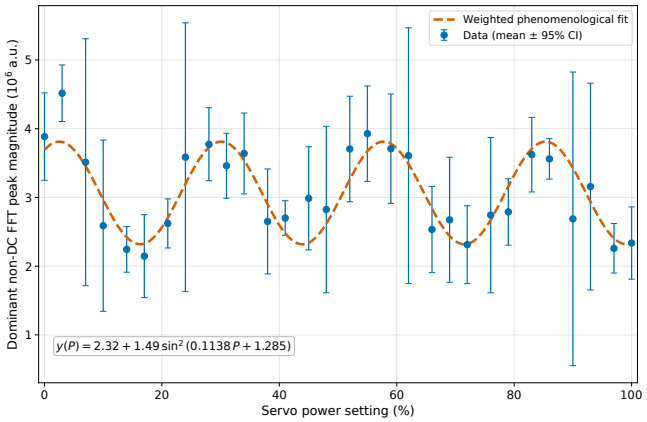


Fig. 4: Dominant non-DC FFT peak magnitude as a function of servo power setting for the power-Rabi scan. At each servo-power setting, five runs were truncated to a common post-transient window beginning at sample 15,000, mean-centered, Hann-windowed, and analyzed with a real FFT after excluding the DC component. Points show the magnitude of the dominant non-DC spectral peak from the run-averaged spectrum, reported in 10^6 arbitrary units for readability. Error bars denote 95% confidence intervals computed from the run-to-run spread in magnitude at the same peak bin, with $n = 5$ runs per power setting. The dashed curve shows a weighted phenomenological fit of the form $y(P) = y_0 + A\sin^2(kP+\phi)$.

store or a consumer-facing online marketplace. The only specialized components not available off the shelf are the PCBs designed for the project. These can either be ordered from a specialty manufacturer or substituted with either a solderless or solderable breadboard. We note that while our initial design made use of a solderable breadboard, the device suffered from considerable electromagnetic noise contributing to the decision to move to a PCB.

Both our initial build and an initial trial build performed by a group of undergraduate seniors in a field trial totaled to a final unit cost of under $500 USD. When compared to the retail price other NMR spectroscopy and computing devices on the market, Quorum-1's total component cost comes in an order of magnitude lower in price (Table I). It should be noted that this does not figure the value of labor put into construction of the device or any profit margin. An itemized bill of materials can be found in appendix Table II.

D. Educational Uses

Quorum-1 is intended to support two complementary modes of undergraduate learning: learning by building quantum-relevant instrumentation and learning by using that instrumentation to explore introductory quantum information science concepts. In the first mode, the device functions as a project-based hardware platform. Students assembling the spectrometer encounter many of the practical skills used across quantum technologies, including analog electronics, soldering, coil construction, signal acquisition, embedded programming, pulse generation, and experimental debugging [8], [34]. These skills are often hidden from students who encounter quantum computing primarily through simulators or cloud-accessible processors, whereas instructional laboratory experiences can expose students to both conceptual and technical aspects of quantum systems [35]. By contrast, a buildable NMR platform exposes the measurement chain directly, requiring students to connect software commands, RF pulses, circuit behavior, and detected spin signals.

In the second mode, Quorum-1 can be used as a laboratory platform for introducing spin-based quantum control. Low-field and Earth-field NMR experiments have previously been used in undergraduate laboratories to teach free induction decay, pulse sequences, relaxation, field homogeneity, and magnetic resonance imaging through hands-on measurements [36]. In Quorum-1, the proton spin in a magnetic field provides a physically accessible two-level system, and the evolution of the measured magnetization can be connected to the Bloch-sphere representation commonly used in quantum information science. Basic experiments such as observing free induction decay, measuring the Larmor frequency, and performing Rabi calibration allow students to connect NMR concepts to coherent rotations, pulse calibration, and single-qubit gate operations. The platform therefore provides a concrete bridge between introductory NMR measurements and the language of gate-based quantum control.

The device also creates opportunities to discuss the limitations of low-field, room-temperature NMR as a quantum-

computing platform. Quorum-1 uses an ensemble of thermally polarized nuclear spins rather than individually initialized qubits, and its measured signal corresponds to ensemble magnetization rather than single-shot projective measurement. These features make the device valuable for teaching spin control, pulse calibration, and measurement, but they also distinguish it from scalable quantum-computing architectures based on individually addressable qubits. This distinction is pedagogically useful: students can compare the strengths of NMR, such as long coherence times and precise pulse control, with its limitations, including weak thermal polarization, ensemble readout, and the absence of scalable initialization and entangling operations in the present device. As a result, Quorum-1 can support not only technical skill development but also critical discussion of what is required for a physical system to function as a quantum computer.

V. Summary

We have developed a low-cost, feature rich single qubit EFNMR device suitable for use in an introductory QIS laboratory or as a build product in an electronics laboratory. Nuclear magnetic resonance spectroscopy makes an ideal platform to investigate topics such as atomic physics, radio frequency engineering, electronics design & building, and quantum state preparation. The skills necessary to build and operate our device are directly transferrable to both technical and research jobs in the QIS industry. We have recently completed a classroom trial of our Quorum-1 kit in an undergraduate senior design course, from which a paper on educational outcomes and lessons learned is forthcoming. Our future work will build from our current device to expand into a two-qubit model suitable for introductory quantum algorithm exploration.

VI. Acknowledgments

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The authors disclose their use of generative AI models ChatGPT-5.4 in the preparation of this manuscript for copy-editing and in troubleshooting electronic and programming issues during project development and Claude Sonnet 4.6 for generating Python code used to generate figures. The authors manually verified all output from AI-generated code.

APPENDIX
BILL OF MATERIALS

The following table is an itemized bill of materials for the construction of a Quorum-1 NMR device as built by the University of Illinois Chicago senior design team from January to April 2026. All prices are in USD.

TABLE II: Bill of materials for one fully assembled Quorum-1 unit.

Component	Qty.	Unit Cost	Total Cost
Polarization Board			
22 Ω resistor	1	$0.06	$0.06
400 Ω resistor	1	$0.21	$0.21
15 kΩ resistor	1	$0.10	$0.10
IRFP4768PBF MOSFET	2	$4.72	$9.44
SIL24-1A72-71D relay	1	$4.08	$4.08
1000 μF polar capacitor	1	$2.55	$2.55
9 V power supply or 9 V battery with solderable 9 V PCB attachment	1	$1.59	$1.59
MUR3060P rectifier	1	$1.98	$1.98
4n46 optocoupler	1	$3.56	$3.56
Right-angle SMA connector	2	$1.96	$3.92
0.5 A fuse	1	$0.35	$0.35
Subtotal			**$27.84**
Main Board			
Raspberry Pi Pico W	1	$8.00	$8.00
330 Ω resistor	1	$0.10	$0.10
370 Ω resistor	1	$0.16	$0.16
1 kΩ resistor	3	$0.10	$0.30
1.5 kΩ resistor	1	$0.10	$0.10
2 kΩ resistor	5	$0.04	$0.20
5 kΩ resistor	1	$0.06	$0.06
10 kΩ resistor	1	$0.06	$0.06
12 kΩ resistor	1	$0.06	$0.06
15 kΩ resistor	2	$0.10	$0.20
47 kΩ resistor	3	$0.03	$0.09
91 kΩ resistor	1	$0.10	$0.10
150 kΩ resistor	1	$0.10	$0.10
330 kΩ resistor	1	$0.10	$0.10
LF411CN op-amp	3	$1.50	$4.50
LT1007 op-amp	3	$6.47	$19.41
1 nF capacitor	2	$0.23	$0.46
10 nF capacitor	4	$0.17	$0.68
18 nF capacitor	1	$0.30	$0.30
100 nF capacitor	2	$0.06	$0.12
MJ-3502N audio jack	2	$1.33	$2.66
SIL24-1A72-71D relay	3	$4.08	$12.24
Diode	2	$0.05	$0.10
Right-angle or vertical SMA connector	2	$1.96	$3.92
Subtotal			**$54.02**
Transmit/Receive Coil			
30 AWG enameled copper wire	1	$52.62	$52.62
3 in DWV pipe	1	$23.83	$23.83
Subtotal			**$76.45**
Polarization Coil			

Continued on next page

Component	Qty.	Unit Cost	Total Cost
18 AWG enameled copper wire	1	$99.57	$99.57
4 in. DWV pipe	1	$21.25	$21.25
Subtotal			**$120.82**
Miscellaneous			
12 V battery	1	$47.99	$47.99
3.5 mm phone plug cable	2	$1.08	$2.16
XT60 connectors with presoldered wire leads	6	$2.50	$15.00
Aluminum shielding core	1	$89.17	$89.17
Electrical tape	1	$2.00	$2.00
9 V battery	1	$1.00	$1.00
Wooden coil assembly stands	1	$20.00	$20.00
Heat-shrink wrap	8	$0.02	$0.16
20 fl oz PET commercial beverage bottle	1	$2.00	$2.00
Subtotal			**$179.48**
Total cost for one fully completed unit			**$458.61**

Component costs reflect approximate retail pricing at the time of assembly.

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